

One symbol, two meanings

Mental picture: two doors, “CLASS wheel” and “ID wheel”. Delta swaps their signs. Following the familiar sign still opens a legal door—but it is the wrong door.

1 The program is legal, but its meaning changes

3DOM class selector

```
(function(){ $S('.wheel').recolor('#111111'); })();
```

delta correct: class selector

```
(function(){ $S('#wheel')#recolor('#111111'); })();
```

delta wrong: id selector

```
(function(){ $S('.wheel')#recolor('#111111'); })();
```

Site: delta:t000:T_CLASS_SIGIL:0 · character offset 17 (zero-based).

prefix = "(function(){ \$S('' correct c = '#' competitor q = '."

In delta, # means CLASS and . means ID. The chain separator is also remapped to #; the color literal #111111 is unchanged. These are DSL strings, not ordinary JavaScript.

$$\text{parse}_\delta(p_c) = \text{parse}_\delta(p_q) = \text{valid}, \quad \text{IR}_\delta(p_c) \neq \text{IR}_\delta(p_q)$$

IR is the canonical meaning: “recolor class(wheel)” versus “recolor id(wheel)”. Hashes 5be0d4314219e5d2... and d74513ab1dcc9490... differ. That is a SEMANTIC collision: silently wrong, not a lexer/parser error.

2 Arm A: weigh the two spellings before choosing

Think of a balance scale. r is the remote mapping/rule; x is the exact prefix; y is a candidate spelling. Score all its model tokens using natural logarithms:

$$\ell_\theta(y \mid r, x) = \sum_{j=1}^{K_y} \log P_\theta(y_j \mid r, x, y_{<j})$$

$$M_\theta^{\text{seq}} = \ell_\theta(c \mid r, x) - \ell_\theta(q \mid r, x) = \log \frac{P_\theta(c \mid r, x)}{P_\theta(q \mid r, x)}$$

$M > 0$ favors the correct spelling; $M < 0$ favors the competitor. One character need not be one tokenizer token. Preference at this forced prefix is not yet proof of what free generation will emit.

$$\widetilde{M} = (-1.0295) - (+3.5227) = -4.5522, \quad \mathbf{1}[\widetilde{M} < 0] = 1$$

Important: the supplied values are planted FakeLM scores, not normalized log probabilities. A genuine log probability cannot be positive. The arithmetic illustrates the pipeline; it is not a measured probability or evidence of model bias.

3 Risk: turn the balance into a danger score

$$S^{\text{seq}} = \ell_b(q \mid r, x) - \ell_b(c \mid r, x) = -M_b(r)$$

$$S^{\text{local}} = \ell_b(q \mid r^0, x) - \ell_b(c \mid r^0, x) = -M_b(r^0)$$

$$S^{\text{shift}} = S^{\text{seq}} - S^{\text{local}}, \quad 4.5522 = 4.6957 + (-0.1435)$$

b = base checkpoint; r^0 = no-rule control (length-matched in real comparisons). Higher S means more competitor preference. Here the rule lowers the synthetic risk by 0.1435, but the balance still favors the wrong spelling.

From this site to the hypotheses

Mental picture: Arm A measures the pull at a junction. Arm B watches the actual journey. H4 predicts dangerous junctions; H5 changes their signs; H2 and H3 ask why the pull occurs.

Arm B: did generation reach the junction and take the wrong door?

O = 1 means the comparable site was reached; Y = 1 means the competitor was emitted there. The supplied fixture has O = 1, Y = 1, parse_valid = True, task_correct = False, n_ops = 1: VALID_WRONG. FakeLM is configured to echo this variant.

$$P(O = 1, Y = 1) = P(O = 1)P(Y = 1 \mid O = 1)$$

Count reach and wrong choice separately. A model that never reaches the junction has not demonstrated that it makes the right choice there.

H4 Prediction: can a warning score identify the dangerous sites?

$$S_i = -M_i^{\text{base}}(r_i), \quad Y_i^A = \mathbf{1}[M_i^{\text{instruct}}(r_i) < 0]$$

$$\text{AUROC} = P(S^+ > S^-) + \frac{1}{2}P(S^+ = S^-)$$

Use base-model risk to predict its instruct sibling's forced-prefix failures (Y^A), paired by site. S^+ is an error-site score; S^- is a correct-site score. AUROC asks how often an error ranks higher, awarding half-credit for ties. Test on unseen mappings/models against length, identity, program NLL and token count; report calibration and error prevalence. One site cannot establish H4.

H5 Repair: replace dangerous signs without changing the destination

$$\text{IR}_{\phi'}(W(p)) = \text{IR}_{\phi}(p) \quad \forall p \in \mathcal{C}$$

$$\eta = \frac{e_{\text{before}} - e_{\text{after}}}{B}$$

W rewrites the program together with its spelling map $\phi \rightarrow \phi'$; C is the tested corpus, e the error rate and B the number of changed symbols. η is benefit per symbol. Targeted repair must beat random repair at the same budget, approach global renaming with fewer changes, and leave low-risk sites stable. Changing # to . alone would change meaning, not repair it.

H2 Interaction: does a local cue amplify a strong old habit?

$$\Delta_Z = (\bar{Z}_{s,h} - \bar{Z}_{s,l}) - (\bar{Z}_{n,h} - \bar{Z}_{n,l})$$

$$\Delta_M < 0, \quad \Delta_{\logit(p)} > 0, \quad \Delta_p > 0, \quad \logit(p) = \log \frac{p}{1-p}$$

s/n = strong/neutral competitor prior; h/l = high/low local cue; p = reversion probability. This four-cell difference-in-differences tests extra amplification, not just “a bad habit exists”. Require a meaningful effect and consistent direction on margin, logit and probability scales. The cue must be independently validated; current Sol notes mark H2 untestable.

H3 Mechanism: a rule pathway and a familiar-completion pathway

$$G_i^{\text{tok}} = z_i(q_i^*) - z_i(c_i^*)$$

$$\Delta_{\text{local}} = G(\text{high} \rightarrow \text{low patched}) - G(\text{low}) > 0$$

$$\Delta_{\text{rule}} = G(\text{correct rule} \rightarrow \text{absent patched}) - G(\text{absent}) < 0$$

z is a token logit; q^* and c^* are the matched next-token alternatives. These are two separate donor $\rightarrow$ recipient patch tests. The prompt shift alone proves no causal pathway. H3 here follows the literature review's definition.

Bottom line: this FakeLM example verifies scoring arithmetic and silent-error classification—not H2/H3/H4/H5. Real held-out tests are required; clean transfer to a genuinely new grammar family is not yet established.